Lesson 6 Container Daily Management

Nowadays, Docker has become one of the core tools for developing and deploying applications. However, to ensure the stability and reliability of the container environment, daily maintenance and troubleshooting are indispensable tasks. This text will introduce some key Docker container maintenance and troubleshooting skills to help you tackle various common issues.

1. Unnecessary Container and Image Regular Cleaning

Over time, Docker master's valuable disk space may be occupied by a large number of unnecessary containers and images. Regularly cleaning up them is the first step in maintaining a healthy container environment.

Press "Ctrl+Alt+T" to open the command line terminal, and enter "docker container prune". Press "Enter" and input "y" to start removing unnecessary stopped containers.

```
pi@raspberrypi:~ $ docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
46de34b3ffb779928582ffc909c13dd9568aff1bcbe2d56099d1029419277f76
```

Press "Ctrl+Alt+T" to open the command line terminal, and enter "docker image prune". Press "Enter" and input "y" to start removing unnecessary images.

```
pi@raspberrypi:~ $ docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
```

2. Monitor Container Performance

It is important to monitor the performance of the containers to ensure their normal operation. The real-time CPU, memory, network and disk usage of containers can be checked with Docker's built-in "stats" command.

Press "Ctrl+Alt+T" to open the command line terminal, and enter "docker stats 0cfc6a98a68c". Press "Enter" to check the real-time performance of the container.

```
pi@raspberrypi:~ $ docker stats 0cfc6a98a68c
```

```
CONTAINER ID   NAME           CPU %   MEM USAGE / LIMIT   MEM %   NET I/O       BLOCK I/O   PIDS
0cfc6a98a68c   jolly_edison   0.00%   0B / 0B             0.00%   4.98kB / 0B   0B / 0B     1
```

3. Check Container Log

Logging container activities is extremely important for troubleshooting and diagnosing issues. Use the command "docker logs" to check the standard output logs of containers.

Press "Ctrl+Alt+T" to open the command line terminal, and enter "docker logs 0cfc6a98a68c". Press "Enter" to check the standard output log of the container "0cfc6a98a68c".

```
pi@raspberrypi:~ $ docker logs 0cfc6a98a68c
root@0cfc6a98a68c:/# exit
exit
pi@raspberrypi:~ $
```

4. Restart Container

Containers may encounter various issues during runtime, including application collapsing and resource depletion. In such cases, restarting the container can be an effective way to resolve the problem.

Press “Ctrl+Alt+T” to open the command line terminal, and enter “docker restart 0cfc6a98a68c”. Press “Enter” to restart the container.

```
pi@raspberrypi:~ $ docker restart 0cfc6a98a68c
```

5. Process Container Network Problem

The network problem between containers may cause communication failure among applications. Use the “docker network” command to manage the container network to check the containers’ IP addresses and port mappings.

Press “Ctrl+Alt+T” to open the command line terminal, and enter “docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' 0cfc6a98a68c”. Press “Enter” to check the network information of the container.

```
pi@raspberrypi:~ $ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' 0cfc6a98a68c
172.17.0.2
```

6. Process Container Storage Problem

Storage problems in containers may involve disk space exhaustion and data loss. Use data volumes and store drivers to better manage container storage.

Press “Ctrl+Alt+T” to open the command line terminal, and enter “docker volume create my-data”. Press “Enter” to create a data volume.

```
pi@raspberrypi:~ $ docker volume create my-data
my-data
```

After successfully creating the data volume, you can mount it to a container to enable data sharing.

```
pi@raspberrypi:~ $ docker run -v my-data:/path/in/container debian
```

Summary

It is crucial to maintain and troubleshoot Docker containers to ensure the stability and reliability of the container environment. Regularly cleaning up unnecessary resources, monitoring container performance, checking container logs, accessing running containers, restarting containers, building custom images, backing up and restoring container data, addressing container network and storage issues, and utilizing container arrangement tools are all essential skills for better managing and maintaining containerized applications.